

A Telegraph-Equation Price Model with Stochastic Variance and Double-Exponential Jumps: Specification, Validation, and a Structural Null Result

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Abstract

We specify and validate the Dirac–Heston–Jump (DHJ) model, a hybrid transition-density model for foreign exchange that replaces the Black–Scholes diffusion with a Goldstein–Kac telegraph process on a log-price lattice, couples it to a Heston stochastic-variance path, and overlays Kou double-exponential jumps. The telegraph representation gives finite propagation speed and non-Gaussian tails at finite relaxation rate κ , and recovers Black–Scholes exactly as $\kappa \rightarrow \infty$. We give the full continuum and discrete specification, a Monte-Carlo-outer / PDE-inner numerical scheme, and a seven-part validation suite: the model reproduces Garman–Kohlhagen call prices to 0.15% relative error at $N = 1600$ lattice sites, satisfies the martingale condition $\mathbb{E}[S_T] = S_0 e^{(r_d - r_f)T}$ to within 0.003%, preserves that condition under compensated jumps to 0.07%, and exhibits zero negative densities and zero measurable boundary mass loss.

The model was then deployed as a directional forecaster and produced 49–50% accuracy over more than 2,500 logged one-day forecasts. We show this was not a calibration failure but structurally predetermined, and prove it two ways. First, we derive in closed form that the model’s “chiral charge” signal satisfies $\bar{Q}_5(T) = \delta_{cp} e^{-(2\kappa + r_d)T}$ exactly: it is the initial spinor asymmetry times a deterministic scalar, carrying no information from the variance process, the jumps, or the PDE itself. Since δ_{cp} was seeded from RSI, the apparatus computed a damped momentum oscillator; we confirm the identity numerically to six significant figures. Second, a correctly specified risk-neutral model is a martingale by construction, so its directional content is exactly the carry; at a one-day horizon this caps achievable accuracy at 50.28% against a break-even threshold of 51.08% after spread, an expectancy of -0.89 pips per trade. The validation suite that certified the model as correct is the same evidence proving it cannot be directionally profitable at that horizon. We argue this is a general trap in applying derivative-pricing machinery to directional forecasting.

1 Introduction

Option-pricing models are calibrated to reproduce the risk-neutral transition density $p^{\mathbb{Q}}(S_T | S_0)$. Directional trading requires the real-world density $p^{\mathbb{P}}$. The two coincide only up to a market price of risk that pricing models, by design, do not estimate. This paper documents what happens when a carefully validated $\mathbb{Q}$ -measure model is deployed as a $\mathbb{P}$ -measure forecaster, and extracts the general lesson.

The model studied here, Dirac–Heston–Jump (DHJ), was built to address three documented deficiencies of Black–Scholes in FX: Gaussian tails, constant volatility, and continuous paths. The first is addressed by replacing the diffusion with a telegraph (Goldstein–Kac) process, which propagates at finite speed c and relaxes to diffusion at rate 2κ , producing excess kurtosis and skew of order $1/\kappa$; the second by a Heston CIR variance path; the third by a Kou jump overlay. Each ingredient is standard. The contribution is the combination, the numerical treatment required to make it stable, and the negative result with its structural explanation.

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2 Model Specification

2.1 State variables

Let $x = \log(S/S_0)$ be the log-price relative to spot and τ calendar time in years. The state is a two-component real field $\psi = (\psi_+, \psi_-)$ on x , where ψ_+ is right-moving probability flux and ψ_- is left-moving flux. Both are real and non-negative: this is a Euclidean (imaginary-time) formulation, so the components are genuine densities, not quantum amplitudes. We define

$$P(x, \tau) = \psi_+ + \psi_-, \quad Q_5(x, \tau) = \psi_+ - \psi_-, \quad (1)$$

the total density and the *chiral density*. The aggregate chiral charge is $\bar{Q}_5(\tau) = \int Q_5(x, \tau) dx$.

2.2 Telegraph dynamics

The field evolves under the coupled system

$$\partial_\tau \psi_+ = D \partial_x^2 \psi_+ - c \partial_x \psi_+ - \nu \partial_x \psi_+ - \kappa \psi_+ + \gamma \psi_- - r_d \psi_+, \quad (2)$$

$$\partial_\tau \psi_- = D \partial_x^2 \psi_- + c \partial_x \psi_- - \nu \partial_x \psi_- - \kappa \psi_- + \gamma \psi_+ - r_d \psi_-, \quad (3)$$

with D the diffusion coefficient, c the propagation speed, κ the flip (relaxation) rate, γ the chiral mixing rate, ν the log-price drift, and r_d the domestic rate entering as discounting. Setting $\gamma = \kappa$ conserves total probability up to discounting. In (P, Q_5) variables, (2)–(3) become

$$\partial_\tau P = D \partial_x^2 P - c \partial_x Q_5 - \nu \partial_x P - r_d P, \quad (4)$$

$$\partial_\tau Q_5 = D \partial_x^2 Q_5 - c \partial_x P - \nu \partial_x Q_5 - 2\kappa Q_5 - r_d Q_5. \quad (5)$$

For $D = \nu = r_d = 0$, eliminating Q_5 gives the classical telegraph equation

$$\partial_\tau^2 P + 2\kappa \partial_\tau P = c^2 \partial_x^2 P, \quad (6)$$

which interpolates between ballistic propagation at speed c (small κ) and diffusion (large κ).

2.3 The Black–Scholes limit and the finite- κ correction

As $\kappa \rightarrow \infty$ the chiral mode relaxes instantaneously. Setting $\partial_\tau Q_5 \approx 0$ and dropping terms small against $2\kappa Q_5$ in (5) gives $Q_5 \approx -(c/2\kappa) \partial_x P$. Substituting into (4),

$$\partial_\tau P = \left(\underbrace{D + \frac{c^2}{2\kappa}}_{D_{\text{eff}}} \right) \partial_x^2 P - \nu \partial_x P - r_d P. \quad (7)$$

With $D = \sigma^2/2$ and $c = \sigma$ this is the exact Black–Scholes Fokker–Planck equation plus an excess-diffusion term. Since $c^2 = 2D$,

$$D_{\text{eff}} = D(1 + \kappa^{-1}), \quad \sigma_{\text{eff}}^2 = \sigma^2(1 + \kappa^{-1}), \quad (8)$$

so the finite-mass model carries a relative variance excess of exactly $1/\kappa$, vanishing in the diffusive limit. Skewness enters at order $1/\kappa$ through the asymmetric momentum sector. These are the intended non-Gaussian corrections.

2.4 Heston stochastic variance

The instantaneous variance $v(\tau)$ follows a CIR process,

$$dv = \kappa_H(\theta - v) d\tau + \xi\sqrt{v} dW_v, \quad (9)$$

with mean-reversion speed κ_H , long-run variance θ , vol-of-vol ξ , and dW_v correlated with the price shock dW_S at $\rho = \text{corr}(dW_S, dW_v)$. It is integrated by a Milstein scheme with full truncation,

$$v_{n+1} = \max\left(v_n + \kappa_H(\theta - v_n) \Delta\tau + \xi\sqrt{v_n^+} \Delta W + \frac{1}{4}\xi^2[(\Delta W)^2 - \Delta\tau], 0\right), \quad (10)$$

where $v_n^+ = \max(v_n, 0)$ and $\Delta W = z_n\sqrt{\Delta\tau}$ with $z_n \sim \mathcal{N}(0, 1)$. Milstein gives strong order 1.0 against Euler's 0.5.

2.5 Variance decomposition: avoiding a double count

The Heston price shock decomposes into correlated and independent components,

$$dW_S = \rho dW_v + \sqrt{1 - \rho^2} dW_\perp. \quad (11)$$

The correlated part is applied separately as a path-conditional translation (Section 3.4), so the PDE must evolve only the independent part. Using the full v would double-count the variance $\rho^2 v$. The PDE coefficients are therefore

$$D(\tau) = \frac{(1 - \rho^2)v(\tau)}{2}, \quad c(\tau) = \sqrt{(1 - \rho^2)v(\tau)}. \quad (12)$$

2.6 Regime-dependent mass

The relaxation rate is made variance-dependent so that calm markets are more ballistic (trending, low κ) and stressed markets more diffusive,

$$\kappa(v) = \min\left(\kappa_0 + \frac{\kappa_1}{v}, \frac{1}{2\Delta\tau}\right), \quad \gamma = \kappa. \quad (13)$$

The κ_1/v term is singular as $v \rightarrow 0$; the cap is required for stability (Section 3.3).

2.7 Kou jumps and the compensator

Jumps arrive as a Poisson process of intensity λ per year. Each jump Y in log-price has the double-exponential density

$$f_Y(y) = p\eta_+ e^{-\eta_+ y} \mathbf{1}_{\{y>0\}} + (1-p)\eta_- e^{\eta_- y} \mathbf{1}_{\{y<0\}}, \quad \eta_+ > 1, \eta_- > 0. \quad (14)$$

Mean jump sizes are $1/\eta_+$ up and $1/\eta_-$ down; asymmetry $\eta_- < \eta_+$ encodes crash skew. The martingale compensator is

$$\zeta = \mathbb{E}[e^Y - 1] = \frac{p\eta_+}{\eta_+ - 1} + \frac{(1-p)\eta_-}{\eta_- + 1} - 1, \quad (15)$$

and $\lambda\zeta$ is subtracted from the drift. The condition $\eta_+ > 1$ is necessary for $\mathbb{E}[e^Y]$ to exist and is enforced at construction.

2.8 Drift and the Garman–Kohlhagen boundary

Under the FX risk-neutral measure with domestic rate r_d and foreign rate r_f , the carry is $r_d - r_f$ and the log-price drift is

$$\nu(\tau) = (r_d - r_f) - \frac{1}{2}v(\tau) - \lambda\zeta, \quad (16)$$

which enforces $\mathbb{E}[S_T] = S_0 e^{(r_d - r_f)T}$. Setting $r_f = 0$ reduces the model to the equity convention. The reference density and call price against which the model is validated are Garman–Kohlhagen,

$$p_{\text{GK}}(S) = \frac{1}{S\sigma\sqrt{2\pi T}} \exp\left[-\frac{(\log(S/S_0) - (r_d - r_f - \frac{1}{2}\sigma^2)T)^2}{2\sigma^2 T}\right], \quad (17)$$

$$C_{\text{GK}} = S_0 e^{-r_f T} \Phi(d_1) - K e^{-r_d T} \Phi(d_2), \quad d_{1,2} = \frac{\log(S_0/K) + (r_d - r_f \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}. \quad (18)$$

2.9 Nested limits

The specification nests three known models exactly: $\kappa_0 \rightarrow \infty$, $\kappa_1 = \xi = \lambda = \rho = 0$ gives Black–Scholes; $\kappa_0 \rightarrow \infty$, $\kappa_1 = \lambda = 0$ gives Heston; $\xi = \lambda = \rho = 0$ gives the deterministic telegraph model. These limits are the basis of the validation suite.

3 Numerical Scheme

The solver is Monte Carlo on the outer loop (variance paths) and PDE on the inner loop (the spinor field conditional on a variance path).

3.1 Lattice and initial condition

The domain is $x \in [-x_h, x_h]$ with $x_h = w\sigma_0\sqrt{T} + \frac{1}{2}$, w half-widths in standard deviations (default 6), discretised at N sites with spacing $\Delta x = 2x_h/(N - 1)$ and Dirichlet boundaries. The initial condition is a Gaussian of width Δx centred at $x = 0$, split asymmetrically,

$$\psi_{\pm}(x, 0) \propto \theta_{\pm} \exp\left(-\frac{x^2}{2\Delta x^2}\right), \quad \theta_+ = \theta_{ic}, \quad \theta_- = 1 - \theta_{ic}, \quad (19)$$

$$\theta_{ic} = \text{clip}\left(\frac{1}{2} + \frac{1}{2}\delta_{cp}, 0.01, 0.99\right), \quad (20)$$

normalised so $\int P dx = 1$. A one-cell width is the narrowest smearing that yields a smooth density while keeping the artificial initial variance $O(\Delta x^2)$ far below $\sigma^2 T$; a width of two or more cells injects 3–7% excess variance at short horizons.

3.2 Spatial discretisation and integrator

With $f_j = f(x_j)$, the operators are

$$\Delta^2 f_j = \frac{f_{j+1} - 2f_j + f_{j-1}}{\Delta x^2}, \quad \nabla^0 f_j = \frac{f_{j+1} - f_{j-1}}{2\Delta x}. \quad (21)$$

Both the wave term and the drift term use the second-order central difference ∇^0 . First-order upwinding would be stable but introduces numerical diffusion $D_{\text{num}} = c\Delta x/2$, which at realistic Δx is comparable to the physical D and would corrupt the very non-Gaussian corrections the model exists to produce. The Wilson diffusion term already damps the high-wavenumber modes that make central differencing unstable in pure advection. Time integration is classical fourth-order Runge–Kutta on the semi-discrete system.

3.3 Stability

Three constraints are enforced. The CFL condition for the explicit scheme is

$$\Delta\tau \leq 0.45 \frac{\Delta x^2}{2D + c \Delta x}. \quad (22)$$

The global step is set from a stationary worst-case variance. The CIR stationary distribution is Gamma with standard deviation $\sigma_v = \xi \sqrt{\theta/(2\kappa_H)}$, so we take $v_{\text{ref}} = \max(\theta + 2\sigma_v, v_0, \theta)$ and size $\Delta\tau$ from (22) at v_{ref} . When a realised path excursion exceeds v_{ref} , the step is subdivided adaptively into $n_{\text{sub}} = \lceil \Delta\tau/\Delta\tau_{\text{CFL}}(v_i) \rceil$ substeps, capped at 32.

Second, the reaction term -2κ is stiff, with RK4 absolute-stability limit $|2\kappa h| < 2.785$. The cap in (13) at $\kappa \leq 1/(2\Delta\tau)$ gives $|2\kappa h| \leq 1$, a $2.8\times$ margin, while still permitting the large κ_0 needed for Black–Scholes recovery tests.

Third, the drift is clamped at $|\nu| \leq \Delta x/(2\Delta\tau)$ as a sanity bound; with (16) it is never binding in practice.

3.4 Leverage as a grid translation

The correlated shock contributes a path-conditional drift $\rho\sqrt{v} dW_v$. Discretising it as a PDE advection term would impose a CFL constraint scaling with $|\rho|\sqrt{v}$, which at realistic FX parameters dominates (22) and forces a much smaller step. Instead it is applied exactly, as a translation of the field after each step:

$$\Delta x_{\text{lev}} = \rho \sqrt{v_i} z_i \sqrt{\Delta\tau}, \quad (23)$$

implemented by sub-cell linear interpolation. Writing $\Delta x_{\text{lev}}/\Delta x = n_s + \phi$ with $n_s = \lfloor \cdot \rfloor$ and $\phi \in [0, 1)$,

$$\tilde{\psi}_j = (1 - \phi) \psi_{j-n_s} + \phi \psi_{j-n_s-1}. \quad (24)$$

Translation is unconditionally stable and mathematically equivalent to the advection term, removing the constraint entirely. Kou jumps are applied by the same operator, sampling $n \sim \text{Poisson}(\lambda\Delta\tau)$ jumps per step and translating by each Y .

3.5 Monte Carlo accumulation

For each of N_p variance paths the field is evolved to T and accumulated with weight equal to its retained norm $\|\psi^{(k)}\|_1$, rather than normalising each path first:

$$\bar{\psi}_j = \frac{\sum_k \psi_j^{(k)}}{\sum_k \|\psi^{(k)}\|_1}, \quad \|\psi^{(k)}\|_1 = \sum_j (\psi_{+,j}^{(k)} + \psi_{-,j}^{(k)}) \Delta x. \quad (25)$$

Per-path normalisation divides by a near-zero norm on paths that have leaked off the domain, amplifying numerical noise catastrophically; weight-based accumulation lets such paths contribute in proportion to the mass they retain. Since total mass decays as $e^{-r_d T}$ under discounting, boundary leakage is diagnosed as

$$\text{mass loss} = \max\left(0, 1 - \frac{\int \bar{P} dx}{e^{-r_d T}}\right). \quad (26)$$

4 Validation

All results below were regenerated from the current build. Test 1 fixes $\xi = \lambda = \rho = 0$ and $\kappa_0 = 500$, where the model must reproduce Garman–Kohlhagen exactly, and refines the lattice (Table 1).

The relative call-price error falls by roughly a factor of three per doubling of N over the first two refinements and reaches 0.15% at $N = 1600$.

N	$\mathbb{E}[S_T]$	GK forward	C_{DHJ}	C_{GK}	rel. err.
200	1.001063	1.001042	0.010286	0.009935	3.54 %
400	1.001048	1.001042	0.010027	0.009935	0.93 %
800	1.001044	1.001042	0.009965	0.009935	0.30 %
1600	1.001043	1.001042	0.009950	0.009935	0.15 %

Table 1: Garman–Kohlhagen recovery against lattice resolution. $S_0 = 1$, $\sigma = 0.082$, $r_d = 0.0525$, $r_f = 0.0400$, $T = 1/12$.

Test 2 sweeps $\kappa_0 \in \{10, \dots, 1000\}$ at $N = 400$ and confirms the limit (7): relative error falls monotonically $3.29\% \rightarrow 1.68\% \rightarrow 1.29\% \rightarrow 1.07\% \rightarrow 0.93\% \rightarrow 0.89\%$, with zero negative density cells throughout. The residual at large κ_0 is the $N = 400$ discretisation floor from Table 1, not a limit failure.

Test 3 checks the martingale condition across horizons and vol-of-vol. Relative deviations of $\mathbb{E}[S_T]$ from $S_0 e^{(r_d - r_f)T}$ are 0.003% ($T = 30\text{d}$, $\xi = 0$), 0.002% (30d , $\xi = 0.3$), 0.016% (91d , $\xi = 0.3$) and 0.001% (30d , $\kappa_0 = 50$). Test 4 repeats this with jumps active at $\lambda \in \{1, 3, 5, 10\}$, verifying the compensator (15): errors are 0.015%, 0.071%, 0.014%, 0.019%, all within Monte Carlo error at $N_p = 300$.

Test 5 finds zero negative density cells across eight configurations spanning $N \in [200, 800]$, $T \in \{1, 3\}$ months, high ξ and λ , and $\rho = -0.6$. Test 6 measures boundary mass loss at 0.0000% for all four standard configurations. The model is, as a numerical implementation of its own specification, correct.

5 Signal Extraction, and Where It Fails

5.1 The extraction layer

Three quantities were read from the solved density. The probability of finishing above spot,

$$\Pi_{\uparrow} = \int_{S_0}^{\infty} p_{\text{DHJ}}(S) dS \approx \sum_{j: S_j \geq S_0} p_j (S_{j+1} - S_j), \quad (27)$$

the expected move $\mathbb{E}[S_T] - S_0$, and the aggregate chiral charge $\bar{Q}_5(T)$, interpreted as a momentum or sentiment proxy. The initial asymmetry was seeded from the 14-period RSI with a MACD agreement filter,

$$\delta_{cp} = \text{clip} \left(\frac{\text{RSI} - 50}{100} \cdot \left(1 - \frac{1}{2} \mathbf{1}_{\{\text{sgn}(\text{RSI} - 50) \neq \text{sgn}(\text{MACD}_h)\}} \right), -\frac{1}{2}, \frac{1}{2} \right), \quad (28)$$

and signals were assigned by thresholds on $\bar{Q}_5$ with confirmation from $\Pi_{\uparrow}$: strong signals at $|\bar{Q}_5| > 0.15$ with $\Pi_{\uparrow}$ confirming beyond 0.52 or 0.48, mild signals at $|\bar{Q}_5| > 0.05$, otherwise neutral.

5.2 The chiral charge is not a model output

Proposition 1. *Under Dirichlet boundaries and $\gamma = \kappa$, the aggregate chiral charge obeys, exactly,*

$$\bar{Q}_5(T) = \delta_{cp} e^{-(2\kappa + r_d)T}. \quad (29)$$

Proof. Integrate (5) over x . The terms $\int \partial_x^2 Q_5 dx$, $\int \partial_x P dx$ and $\int \partial_x Q_5 dx$ are boundary terms and vanish under Dirichlet conditions, leaving $\frac{d}{d\tau} \bar{Q}_5 = -(2\kappa + r_d) \bar{Q}_5$. The leverage and jump operators are translations (24), which preserve the integral, so they do not enter. At $\tau = 0$, by (20), $\bar{Q}_5(0) = \theta_+ - \theta_- = 2\theta_{ic} - 1 = \delta_{cp}$. Integrating gives (29). $\square$

We verified (29) against the compiled engine at $\delta_{cp} \in \{-0.5, \dots, 0.5\}$, $\sigma = 0.082$, $T = 1/365$, $\kappa_0 = 1.0$, $\kappa_1 = 0.002$, giving $\kappa = 1.2974$ and a predicted factor $e^{-(2\kappa + r_d)T} = 0.992773$. The ratio of measured to predicted charge was 1.00000 at every point, to all six digits reported.

The consequence is severe. The chiral charge, which determined the direction of every signal the model emitted, is the RSI-derived seed multiplied by a deterministic scalar within 0.8% of unity at a one-day horizon. It contains no information from the Heston path, the jump process, the leverage coupling, or the PDE evolution. Inverting the thresholds through (28), the signal rule was exactly

$$\text{STRONG} \iff |\text{RSI} - 50| > 15.11, \quad \text{MILD} \iff |\text{RSI} - 50| > 5.04, \quad (30)$$

doubling to 30.2 and 10.1 when MACD disagreed. A lattice PDE with stochastic variance and compensated jumps was computing $\text{RSI} > 65$.

This is a specification error of a particular and instructive kind. Nothing in Section 4 could have caught it: the model was correct, the solver was correct, and the quantity being read out was simply not a function of either. The conserved-quantity analysis that proves (29) takes one page and was not performed until after deployment.

6 Empirical Results

The model ran in production against a live FX feed, logging a forecast per instrument per hourly cycle with a one-day evaluation horizon, scored on realised direction. Accuracy was assessed with the Wilson score lower bound at $z = 1.64$,

$$p_{\text{LB}} = \frac{\hat{p} + \frac{z^2}{2n} - z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad (31)$$

which is preferred to the normal interval because it does not degenerate at small n or extreme $\hat{p}$: a cell at 4/5 carries $p_{\text{LB}} \approx 0.38$ and is correctly not mistaken for an edge.

Over more than 2,500 evaluated one-day forecasts the model achieved 49–50% directional accuracy, indistinguishable from a coin flip. A conditional analysis partitioned forecasts into cells by signal strength, volatility regime, session and instrument. Several cells showed in-sample accuracy of 66–77%; on out-of-sample data the same cells realised 15–31%. The sign reversal, rather than mere decay toward 50%, is the signature of fitting noise across a large cell count, and is what motivated adopting pre-registered interpretation rules for all subsequent tests. An independent five-voter ensemble built to replace the model scored 49.8% on $n = 325$ out-of-sample forecasts, $p_{\text{LB}} = 0.453$.

As a positive control, the same evaluation harness applied to index buy-and-hold recovered a Sharpe ratio of 0.55, confirming that the detector was capable of finding a real effect where one exists.

7 Why the Null Result Was Structural

Two arguments, one specific and one general, show the outcome was not a calibration failure.

The specific argument is Proposition 1: the directional signal was an RSI threshold rule, and momentum rules of that form in liquid FX have been repeatedly documented to carry no exploitable edge net of costs.

The general argument is that the model was, by construction, a martingale. Test 3 confirms $\mathbb{E}[S_T] = S_0 e^{(r_d - r_f)T}$ to 0.003%. Its entire directional content is therefore the carry, and nothing else. Under the model’s own log-normal $\mathbb{Q}$ -density, the implied probability of an up move is

$$\Pi_{\uparrow}^{\mathbb{Q}} = \Phi\left(\frac{m}{s}\right), \quad m = (r_d - r_f - \tfrac{1}{2}\sigma^2)T, \quad s = \sigma\sqrt{T}. \quad (32)$$

Against a symmetric entry with a one-standard-deviation target and stop $d = S_0\sigma\sqrt{T}$ and round-trip cost C , expectancy is $d(2\Pi_{\uparrow} - 1) - C$, so the break-even accuracy is

$$\Pi^* = \tfrac{1}{2} + \frac{C}{2d}. \quad (33)$$

Table 2 evaluates both for EUR/USD at $\sigma = 0.082$, $r_d - r_f = 125$ bp, $S_0 = 1.08$, $C = 1.2$ pips.

Horizon	$\Pi_{\uparrow}^{\mathbb{Q}}$	Π^* (break-even)	edge (pp)	$\mathbb{E}[\text{pips}]$
1 day	0.5028	0.5108	−0.80	−0.89
1 week	0.5063	0.5048	+0.15	+0.36
1 month	0.5128	0.5023	+1.05	+5.36
3 months	0.5222	0.5014	+2.09	+18.48
1 year	0.5444	0.5007	+4.37	+77.38

Table 2: Achievable directional accuracy under the model’s own risk-neutral density versus the accuracy required to break even after spread.

At the one-day horizon at which the model was deployed, its own maximum attainable accuracy of 50.28% falls 0.80 percentage points short of the 51.08% required to cover the spread, an expectancy of −0.89 pips per trade. The observed 49–50% is fully consistent with this bound. No improvement in calibration, lattice resolution or jump parameters could have changed it, because the deficit is a property of the risk-neutral specification rather than of its numerical solution. The result was derivable in closed form before a single forecast was logged, from quantities the validation suite was already reporting.

Table 2 also shows the diagnosis. The gross edge turns positive beyond roughly one week and grows with horizon, because the carry signal accumulates linearly in T while noise grows as $\sqrt{T}$ and the spread is paid once. The model was not wrong; it was deployed at the one horizon where its signal is structurally smaller than its transaction cost. That positive gross expectancy should not be read as free money: at longer horizons this is simply the carry trade, whose returns are widely understood as compensation for crash risk. Our own backtests over 2014–2026 found time-series carry at Sharpe 0.00 and cross-sectional carry at 0.14 ± 0.26 , consistent with a risk premium that is real but not large relative to its variance.

8 Discussion

Three lessons generalise beyond this model.

Validation proves consistency, not usefulness. Every test in Section 4 asks whether the implementation matches the specification. None asks whether the specification answers the question being

posed. A model can pass a complete numerical validation suite and still be structurally incapable of the task it was built for, and ours did.

Risk-neutral machinery does not transfer to directional forecasting. The martingale property is the defining virtue of a pricing model and a disqualifying defect in a forecaster. Practitioners moving pricing code into a signal pipeline should state the measure change explicitly and estimate it, rather than reading $\mathbb{P}$ -measure conclusions off a $\mathbb{Q}$ -measure density.

Conserved quantities should be checked before deployment, not after. Proposition 1 is elementary. Had the integral of (5) been taken at specification time, the entire signal layer would have been recognised as a reparameterised RSI before any code was written.

The model retains genuine value in the role it was designed for. It produces a validated, arbitrage-consistent transition density with tunable tails and skew, it prices European options against a Garman–Kohlhagen benchmark to sub-percent accuracy, and the parameters κ , ξ , ρ and λ give independent control over excess kurtosis and skew for volatility-surface work. The natural next application is variance forecasting rather than direction: returns in liquid FX appear unforecastable at short horizons, but realised variance is strongly persistent, and the model’s variance layer is the component the validation suite most directly certified.

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